Trash Talk

Understanding Memory Management

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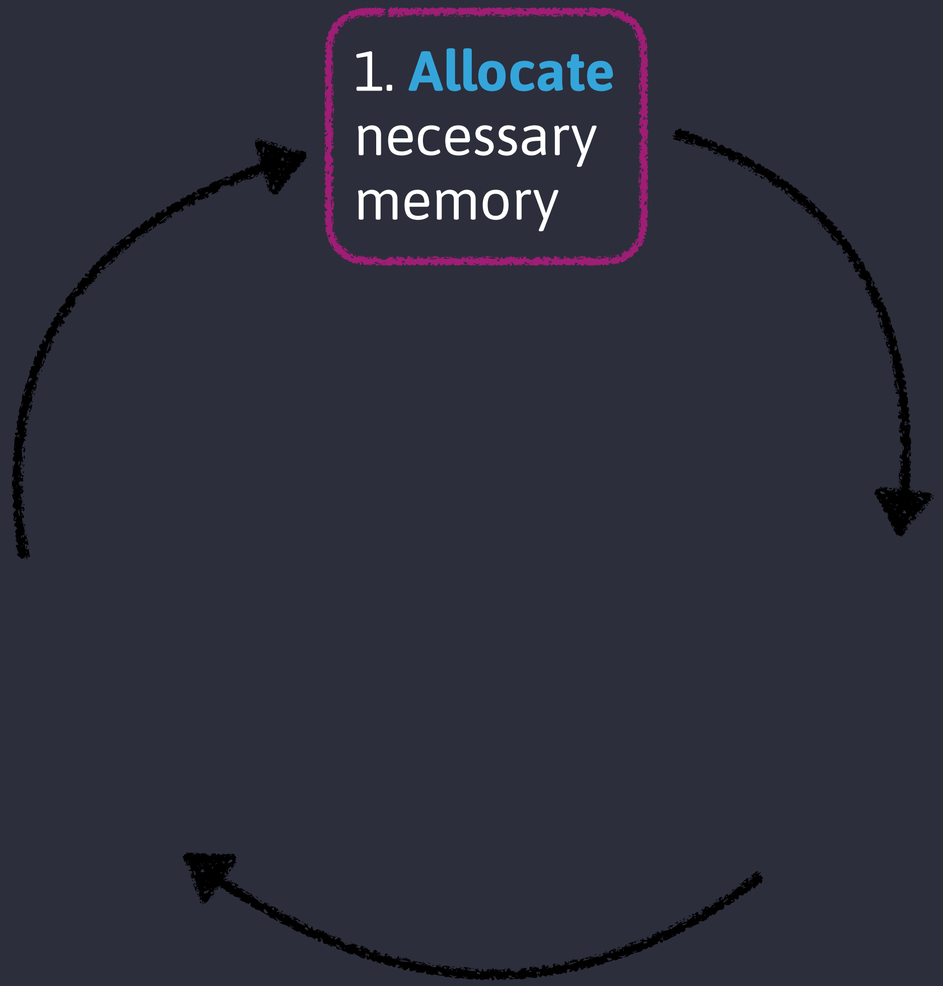
Garbage Collection

¯_(•.•)_/¯ You should probably care about memory management in JavaScript

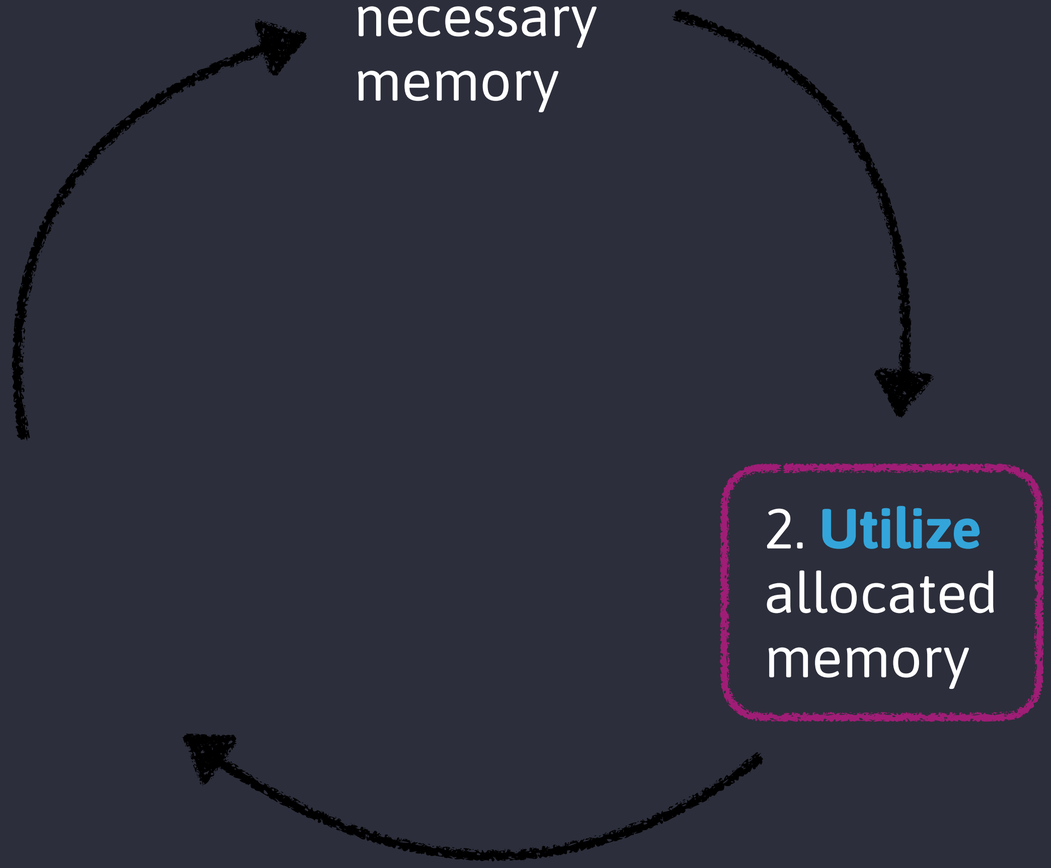
A form of **automatic memory management.** Attempts to **reclaim garbage,** or memory occupied by objects that are no longer in use by the program.

Memory Lifecycle



Explicit in lower-level languages like C, **implicit** in JavaScript

Memory Lifecycle



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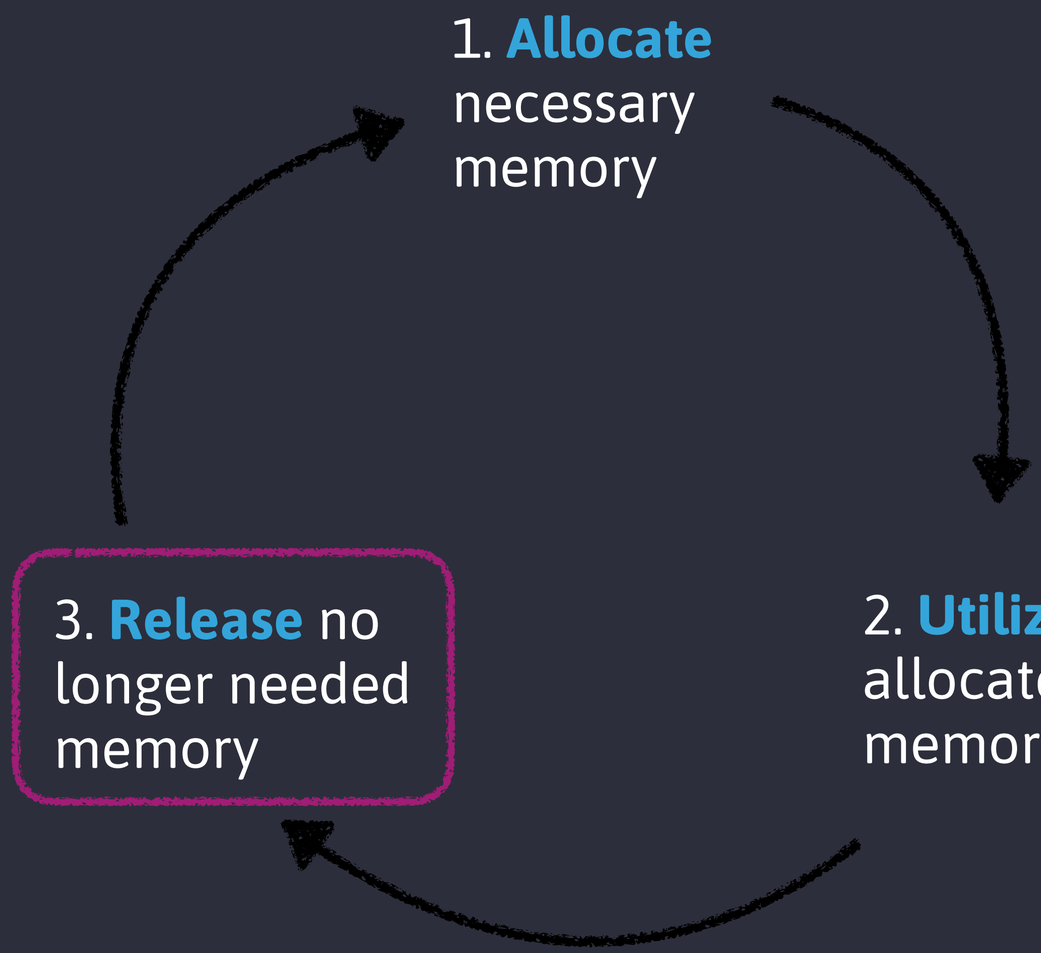
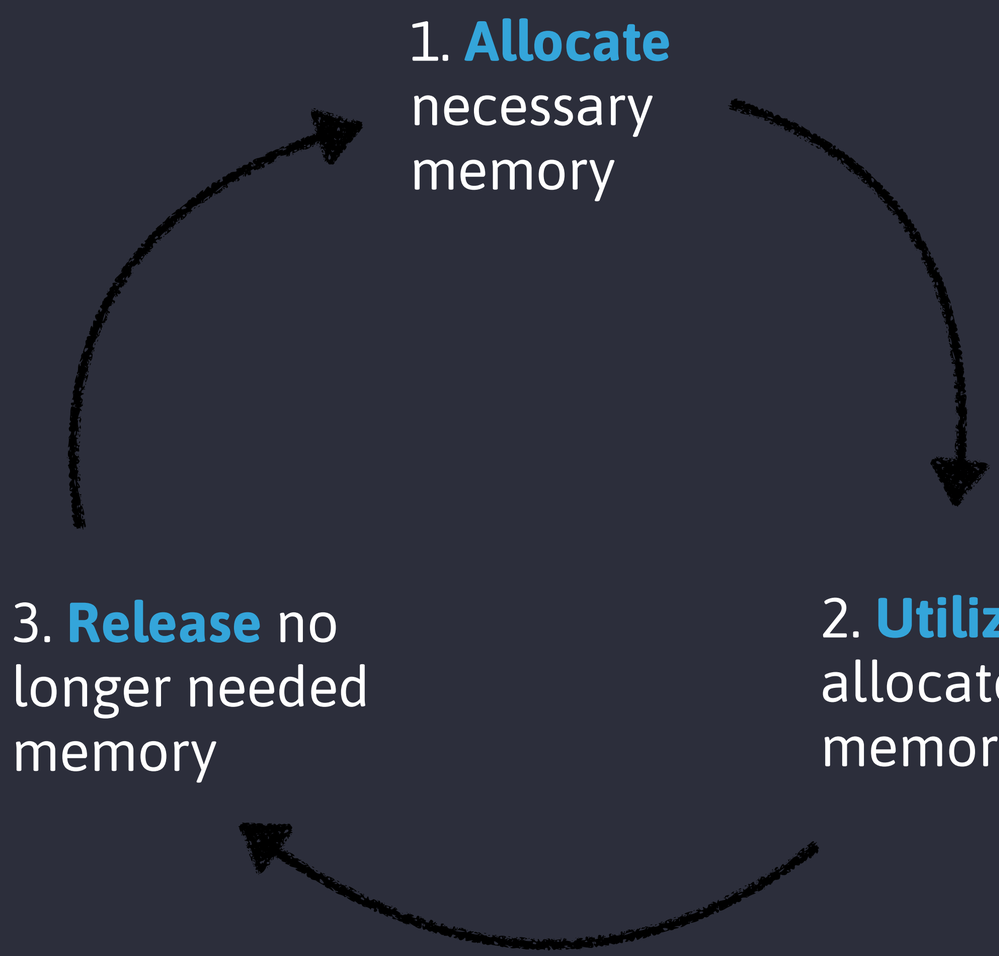
Memory Lifecycle



Explicit in lower-level languages like C, **implicit** in JavaScript

Memory Lifecycle



Explicit in lower-level languages like C, **implicit** in JavaScript

1. Allocate Necessary Memory

Value Initialization

```
// Allocate memory for a number
const num = 42

// Allocate memory for an object
let name = {
  first: 'Shelley',
  last: 'Vohr'
}

// Allocate memory for an array
const colors = ['red', 'green', 'blue']
```

Allocation Via Function Call

```
// Allocate memory via function call
function greet() {
  console.log('Hello, world!');
}
```

2. Use Allocated Memory

Reading and **writing**
in allocated memory

```
let num = 42
num += 10

const person = { firstName: 'Caroline' }
person.firstName = 'Shelley'

let colors = ['red', 'green', 'blue']
colors.push('yellow')
```

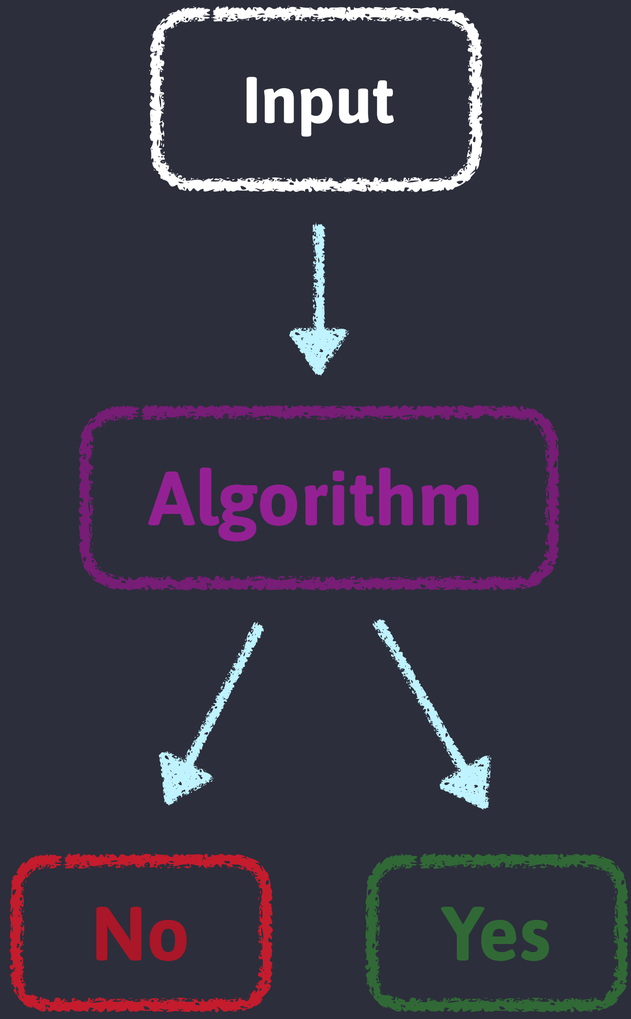
3. Release No Longer Needed Memory

Determining whether memory is reclaimable is an **undecidable problem.**

How do we know when an allocated piece of memory is **garbage collectible**?

Decidability

A problem is **decidable** when a proven effective method for **determining membership** exists for all cases.



Reference-Counting

Whether or not an object is **still needed** → whether an object still has any **other objects referencing it**

Circular references

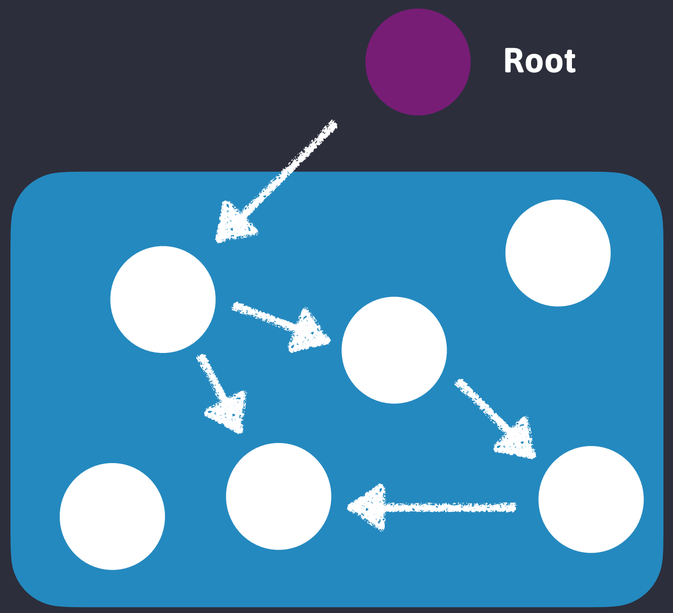
Performance overhead

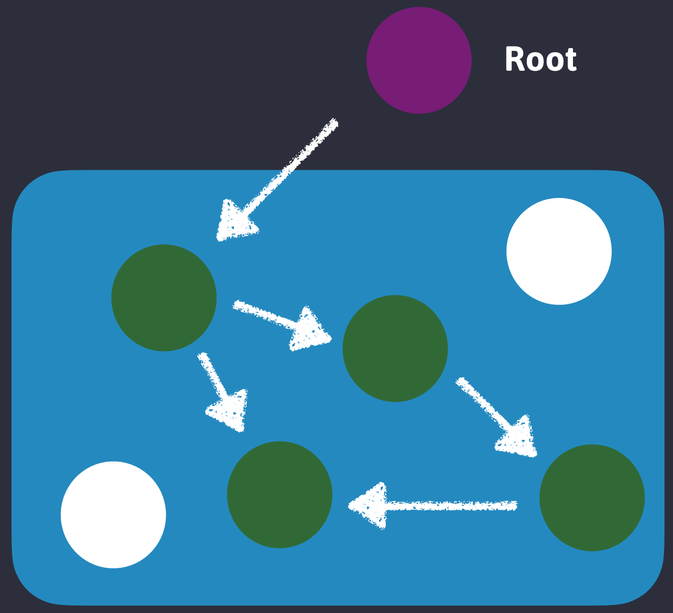
```
// 2 objects created; nothing can be gc'd
const first = { a: { b: 'covalence' } }

const second = first

first = 1

const third = second.a

// 'first' can now be gc'd, but 'a' is
// still referenced and can't be
second = 'electron'

// 'a' can now be gc'd
third = null
```

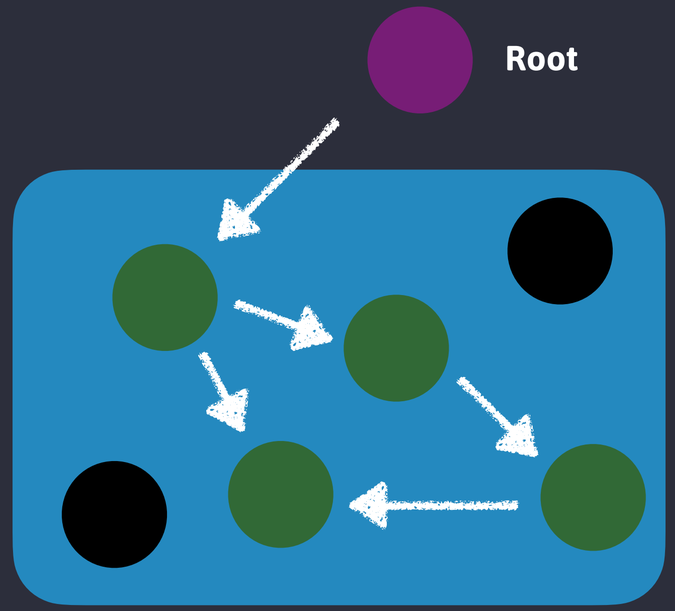
Mark-and-Sweep

Whether or not an object is **still needed** → whether an object is **unreachable**

Can't manually release memory

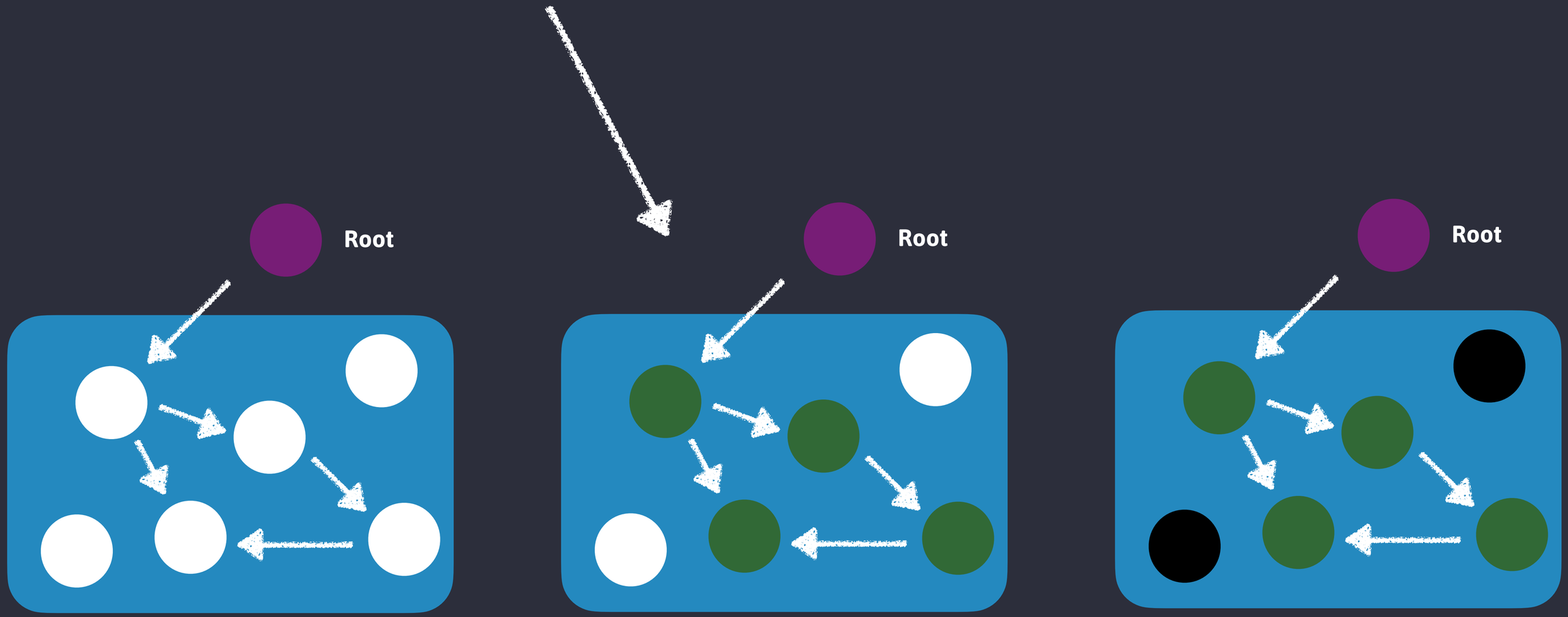






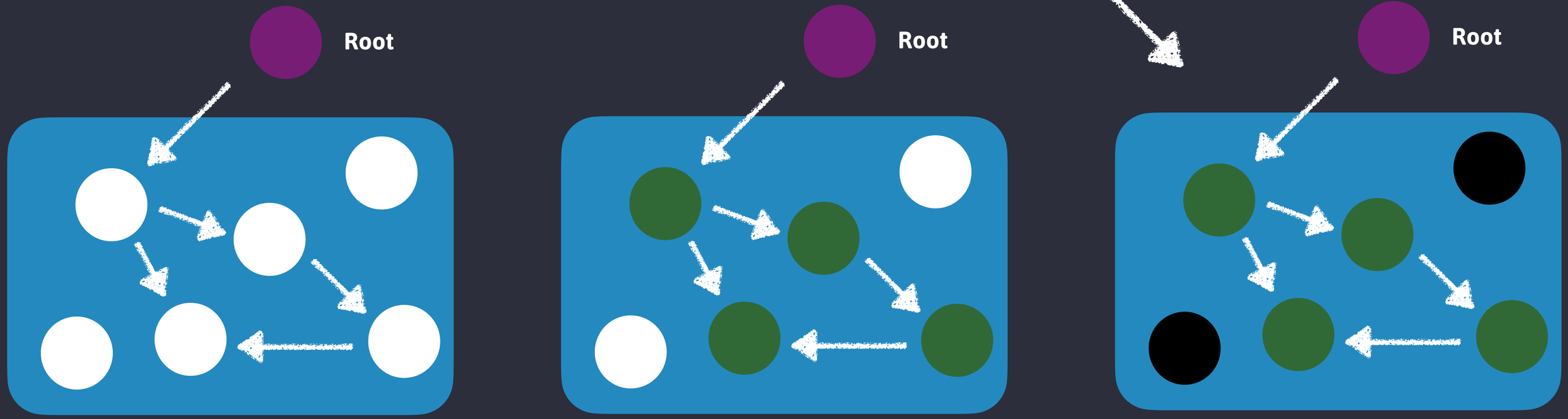
Marking

The process by which **reachable objects** are **found.**



Sweeping

The process by which **unreachable objects** are **cleared** from **heap memory.**



V8

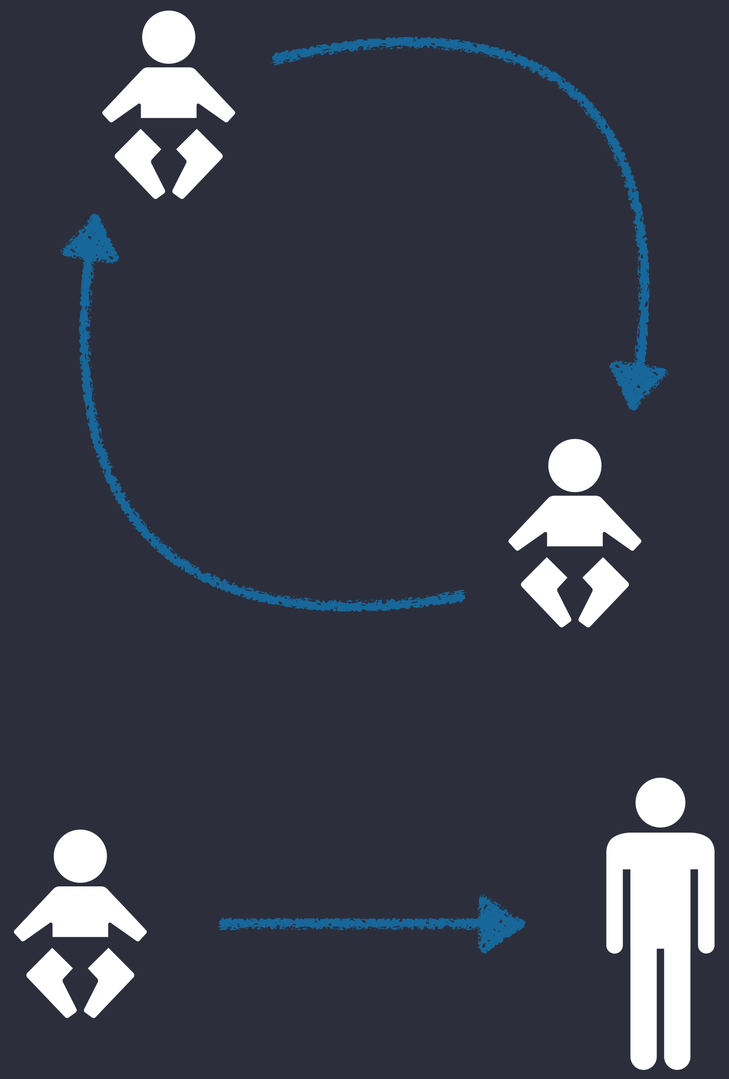
Google's open source **high-performance** JavaScript and WebAssembly **engine**, written in C++

Contains a **mark-and-sweep garbage collector**.

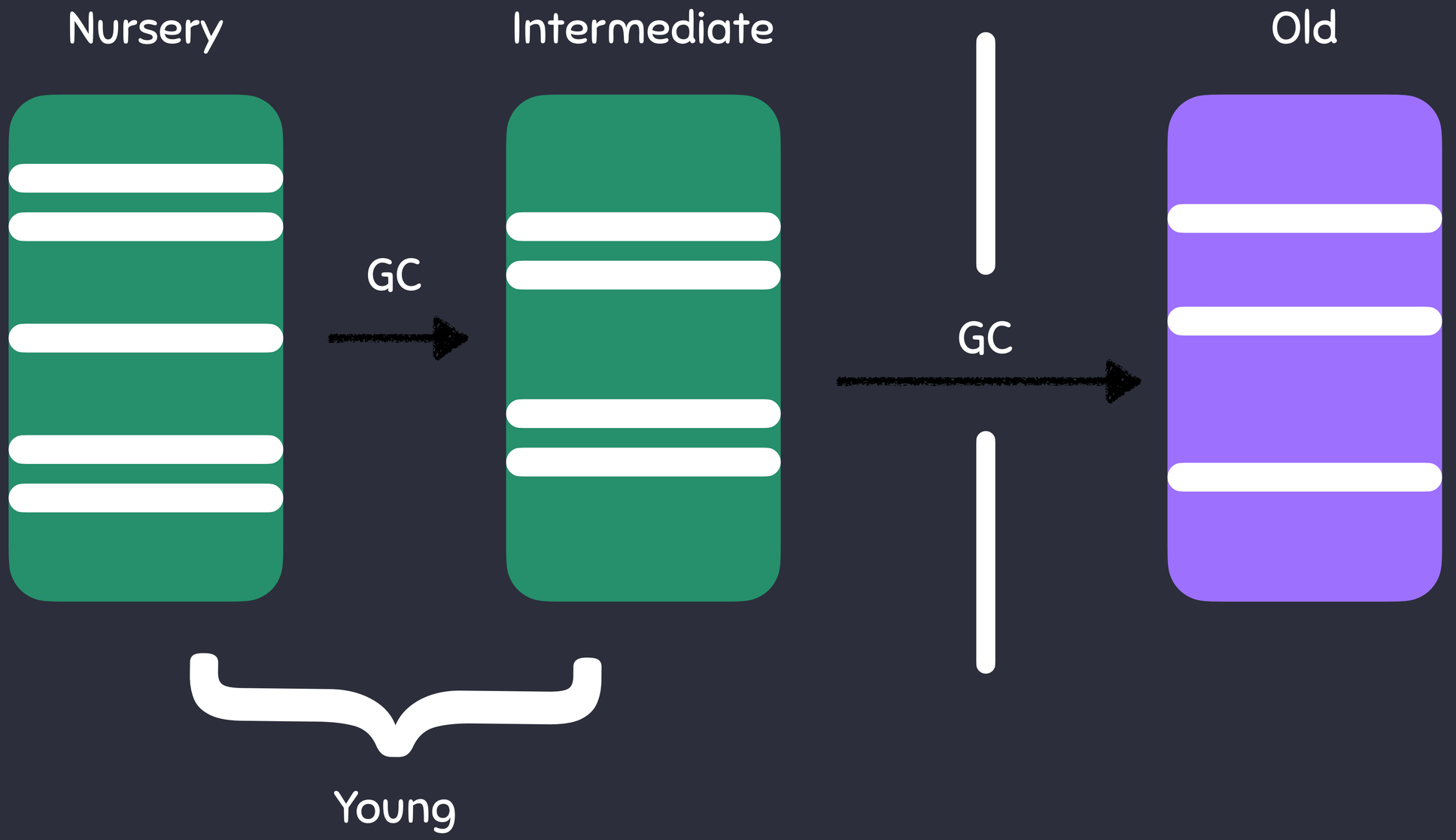
Generational Hypothesis

– Most allocated values become **unused quickly**

– The ones that don't, usually **survive** for a long time



Heap Layout

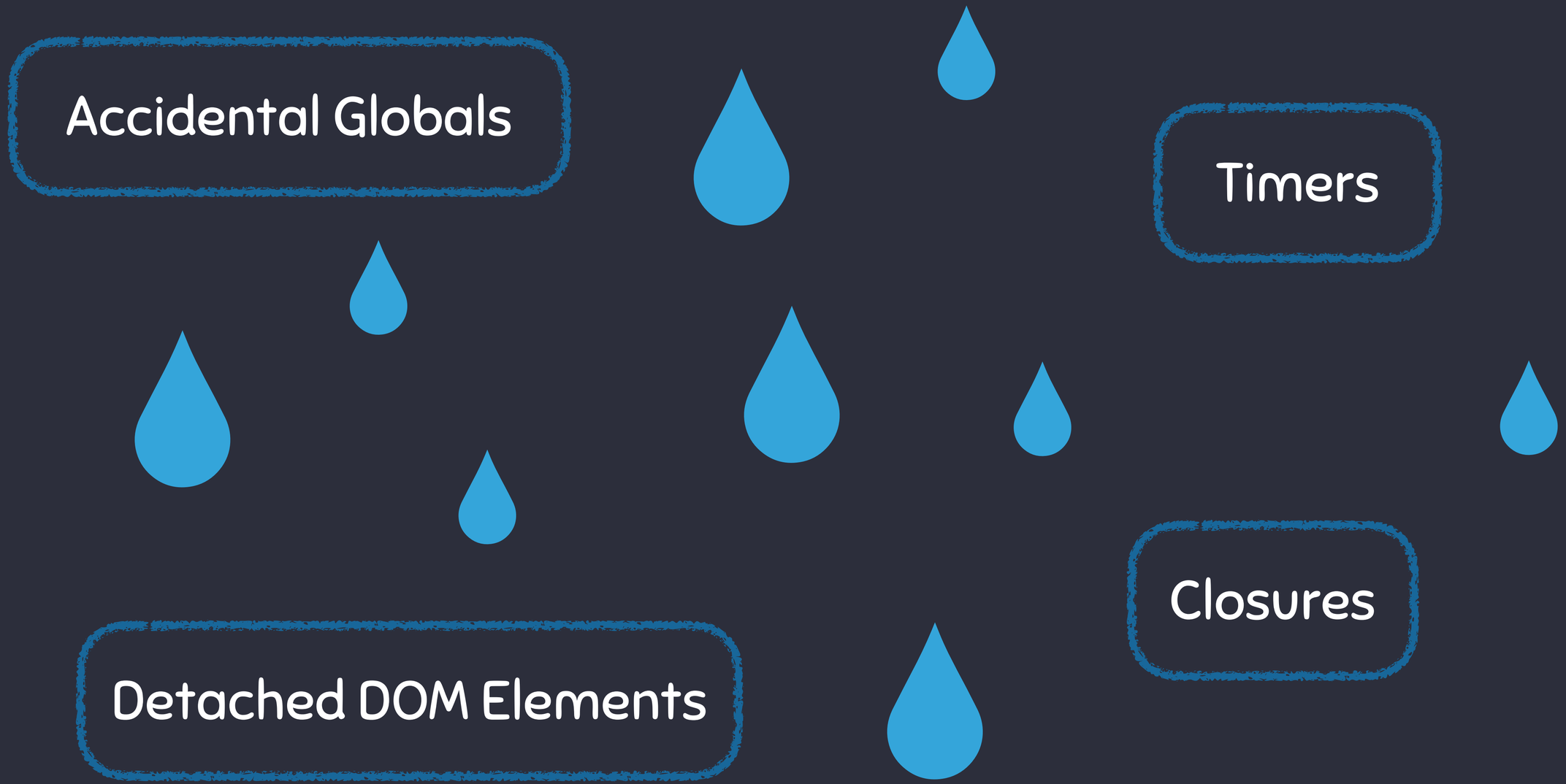


What's a Memory Leak?

A **memory leak** is an allocated piece of memory that the garbage collector can't reclaim

Leads to **increased memory usage** and **degraded performance**

Common Leaks



Accidental Globals

```
function createSession(name, role) {
  sessionData = {}
  sessionData.user = name
  sessionData.role = role
  return sessionData
}
```

Forgot to use let const or var!

`sessionData` will never go out of scope and get garbage collected.

Accidental Globals

```
function createSession(name, role) {
  const sessionData = {}
  sessionData.user = name
  sessionData.role = role
  return sessionData
}
```

Used const!

`sessionData` goes out of scope and gets garbage collected.

Closures

```
function makeClosure () {
    const data = new Array(1000000).fill( 'Data')
    return function () {
        console.log(data)
    };
}

const closure = createClosure()
closure()
closure()
```

Returned function keeps a reference to `data`

Closure is formed over `data`, preventing garbage collection

Closures

```
function makeClosure() {
    let data = new Array(1000000).fill('Data')
    return function () {
        console.log(data)
        data = null
    };
}

const closure = makeClosure()
closure()
closure()
```

Returned function releases `data`

Closure is not formed over `data`, allowing garbage collection

Timers

```
function startTimer() {
    const data = new Array(1000000).fill("Data")
    setInterval (() ⇒ {
        console.log("Timer running")
    }, 1000);
}

startTimer()
```

`setInterval` keeps a reference to `data` as long as the timer is running (forever!)

Memory leakage and buildup if `startTimer` is called multiple times

Timers

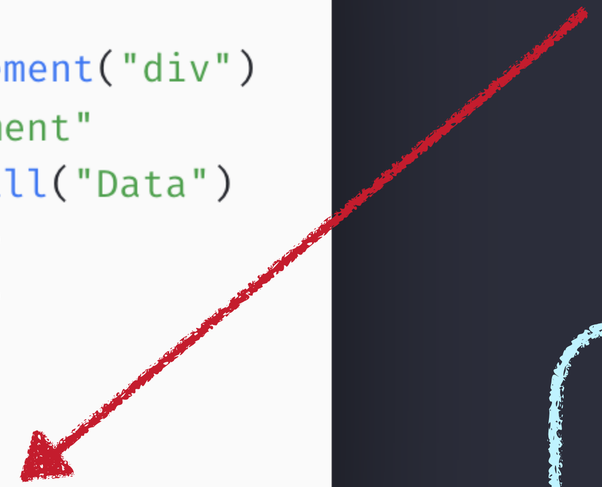
```
function startTimer() {
    const intervalId = setInterval(() => {
        console.log("Timer running")
    }, 1000)

    setTimeout(() => {
        clearInterval(intervalId)
        console.log("Timer stopped")
    }, 5000)
}

startTimer()
```

The timer is cleared when it is no longer necessary!

No memory leakage and buildup if `startTimer` is called multiple times!

Detached DOM Elements

```
const elementMap = new Map()

function createMappedElement () {
    const element = document.createElement("div")
    element.textContent = "Mapped Element"
    const data = new Array(1000000).fill("Data")
    elementMap.set(element, largeData)
    document.body.appendChild(element)
}

createMappedElement()
const element = document.querySelector("div")
document.body.removeChild(element)
```



`element` is later detached from the DOM

JavaScript `Map` objects strongly reference their keys, `element` and `data` remain in memory.

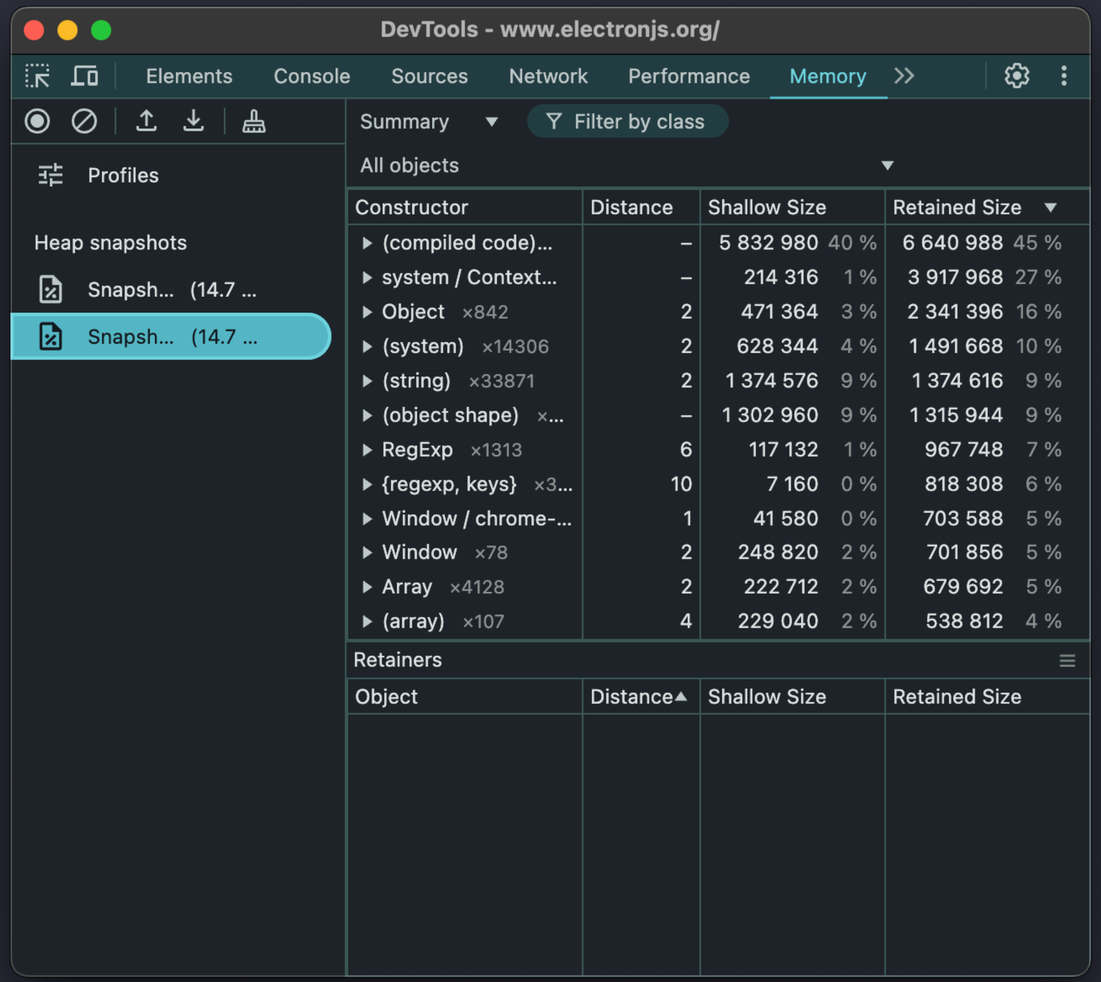
Detached DOM Elements

```
const elementMap = new Map()

function createMappedElement () {
    const element = document.createElement("div")
    element.textContent = "Mapped Element"
    const data = new Array(1000000).fill("Data")
    elementMap.set(element, largeData)
    document.body.appendChild(element)
}

createMappedElement()
const element = document.querySelector("div")
document.body.removeChild(element)
elementMap.delete(element)
```

`element` and `data` are eligible for garbage collection

`element` is removed from `elementMap`

Heap Snapshot



- Understand Memory Usage
- Diagnose Memory Leaks
- Gain Insights
- Optimize Performance

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WeakMap

```
const elementMap = new WeakMap()

function createMappedElement() {
    const element = document.createElement("div")
    element.textContent = "Mapped Element"
    const data = new Array(1000000).fill("Data")
    elementMap.set(element, data)
    document.body.appendChild(element)
}

createMappedElement()

const element = document.querySelector("div")
document.body.removeChild(element)
```

- Keys are objects only
- Automatically removes keys when unreferenced
- Good for caching and private object data

WeakSet

```
let activeUsers = new WeakSet()

let user1 = { name: 'Carolyn' }
let user2 = { name: 'Caroline' }

activeUsers.add(user1)
activeUsers.add(user2)

console.log(activeUsers.has(user1)) // true
console.log(activeUsers.has(user2)) // true

// user1 can now be garbage collected
user1 = null
```

- Holds object references only
- Automatically removes keys when unreferenced
- No iteration or size checking

FinalizationRegistry

An object that lets you run **cleanup logic** when an object is garbage-collected

Avoid or use with caution!

```
const registry = new FinalizationRegistry((heldValue) ⇒ {
  console.log(`Cleaning up resource associated with: ${heldValue}`);
});

let user = { name: "Shelley" };

registry.register(user, "Shelley's resource");

user = null;

console.log("User object is now eligible for garbage collection.");
```

- Unpredictable Timing
- Performance Impact
- Hard to Debug

Wrapping Up

While you can get by in JavaScript with only cursory knowledge of garbage collection, you'll write **more robust code** when you consider it actively!

- ✓ **Actively consider** UI element **lifetimes**
- ✓ **Profile** your **heap usage** regularly
- ✓ **Watch** your **globals**

THANK YOU!

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